



ZooGov: A Full-Stack DAO Operating System for Conservation Research

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March 2025

Abstract

We present **ZooGov**, a full-stack decentralized autonomous organization operating system purpose-built for conservation research governance. ZooGov implements the complete lifecycle of on-chain governance—proposal creation, token-weighted and quadratic voting, vote delegation, committee management, treasury integration, timelock execution, and compliance veto—as a composable smart contract system deployed on Zoo Network’s Lux-based EVM L2. Unlike general-purpose DAO frameworks (Compound Governor, Aragon OSx, Gnosis Safe), ZooGov is specialized for nonprofit research coordination under a 501(c)(3) legal wrapper, incorporating milestone-gated grant disbursement, soulbound reputation weighting, and a narrow board veto mechanism that preserves both community sovereignty and legal compliance. We formalize the proposal type taxonomy from ZIP-0017, prove liveness and safety properties of the governance state machine, and describe the production deployment powering **`governance.zoo.network`**. The reference implementation processes five proposal types with differentiated quorum, approval, and timelock parameters, enabling the DAO to handle routine parameter changes in 9 days and emergency security responses in under 4 hours. All governance parameters are themselves governed through meta-governance proposals, making the system fully self-amending.

Keywords: DAO governance, smart contracts, quadratic voting, delegation, treasury management, conservation science

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1 Introduction

Decentralized autonomous organizations coordinate collective decisions through smart contracts rather than corporate boards. Since the original DAO on Ethereum in 2016, the design space has matured through Compound’s Governor framework [Compound(2020)], Aragon’s modular DAO toolkit [Aragon(2023)], Gnosis Safe’s multi-signature execution [Gnosis(2021)], and Snapshot’s off-chain voting [Snapshot(2021)]. Each advances one aspect of DAO infrastructure but none provides a complete operating system for the specific requirements of nonprofit research governance.

Conservation research DAOs face constraints absent from DeFi governance. First, the 501(c)(3) legal wrapper requires a board compliance veto to maintain tax-exempt status (ZIP-0000 §15). Second, proposal types span orders of magnitude in impact—from a 50 ZOO parameter tweak to a 500,000 ZOO multi-year research grant—demanding differentiated governance parameters. Third, meritocratic input from active researchers must be weighted alongside economic stake, requiring integration with a soulbound reputation system (ZIP-0026). Fourth, grants require milestone-gated disbursement rather than lump-sum treasury transfers.

ZooGov addresses these requirements as an integrated system rather than a collection of loosely coupled contracts. It implements ZIP-0017 (DAO Governance Framework) as the core governance engine, with extension points for ZIP-0023 (Grants), ZIP-0026 (Reputation), and ZIP-0027 (Emergency Governance).

1.1 Contributions

1. A **complete governance contract system** implementing five proposal types with differentiated parameters (Section 4).
2. A **delegation model** with transitive depth limits and reputation-weighted override (Section 5).
3. A **committee framework** for specialized governance bodies (Section 6).
4. **Treasury integration** with multi-chain asset management and milestone escrow (Section 7).
5. **Safety and liveness proofs** for the governance state machine (Section 9).
6. A **production deployment** analysis of the `zoogov-app` frontend and on-chain backend (Section 11).

2 Related Work

2.1 Compound Governor

Compound’s Governor Bravo [Compound(2020)] established the proposal-vote-timelock-execute pattern adopted by hundreds of DeFi protocols. Proposals encode calldata for target contracts; token holders vote; a timelock delays execution. Limitations: single proposal type, no delegation depth control, no reputation integration, no milestone disbursement. OpenZeppelin’s Governor [OpenZeppelin(2023)] modularizes Bravo into composable modules (voting, counting, timelock) but retains the same expressiveness constraints.

2.2 Aragon OSx

Aragon’s OS refactored [Aragon(2023)] provides a plugin architecture where governance logic is installed as plugins on a DAO entity. This enables arbitrary governance configurations but requires significant integration work. ZooGov takes a different approach: a purpose-built system with fixed proposal types and differentiated parameters, trading generality for correctness guarantees.

2.3 Gnosis Safe

Gnosis Safe [Gnosis(2021)] provides multi-signature wallets with modular transaction guards. Safe serves as the treasury execution layer in many DAOs. ZooGov uses Safe-compatible multi-sig for treasury operations but adds governance-controlled spending limits, milestone escrow, and compliance veto.

2.4 Snapshot and Off-Chain Voting

Snapshot [Snapshot(2021)] enables gasless off-chain voting via signed messages. While eliminating gas costs, off-chain votes require a trusted relayer for execution and cannot enforce on-chain timelocks. ZooGov uses on-chain voting for binding proposals and supports Snapshot-compatible signaling for non-binding polls.

2.5 Tally and Governance Frontends

Tally [Tally(2023)] provides a generic governance frontend for Governor-compatible contracts. ZooGov’s frontend (`zoogov-app`) is specialized for research governance, with proposal templates, milestone tracking, grant application flows, and reputation display.

2.6 Science DAOs

VitaDAO [VitaDAO(2022)], LabDAO, and Molecule govern research funding but use standard Governor or Snapshot mechanics without research-specific proposal types, milestone gating, or 501(c)(3) compliance integration.

3 Governance Architecture

3.1 Design Principles

ZooGov follows three principles:

1. **Differentiated governance:** Proposal impact determines quorum, approval threshold, time-lock duration, and required governance bodies.
2. **Dual-token weighting:** Voting power combines transferable ZOO stake with non-transferable ZooRep reputation, preventing pure plutocracy.
3. **Compliance integration:** A narrow board veto operates at the timelock layer, preserving 501(c)(3) status without centralizing decision-making.

Table 1: Proposal types with governance parameters.

Type	Quorum	Approval	Timelock	Scope
Parameter Change	4%	>50%	48h	Protocol tuning
Treasury Spend	10%	>60%	72h	Fund allocation
Protocol Upgrade	15%	>66%	168h	Contract upgrades
Emergency Action	1%	>80%	0h	Security response
Meta-Governance	20%	>75%	168h	Rule changes

3.2 Proposal Type Taxonomy

ZIP-0017 defines five proposal types:

Each type has its own quorum denominator (percentage of total delegated voting power), approval threshold (percentage of non-abstaining votes), and timelock duration.

3.3 Governance State Machine

Definition 3.1 (Proposal States). *A proposal p transitions through the state machine $\mathcal{S} = \{\text{Pending}, \text{Active}, \text{Succeeded}, \text{Defeated}, \text{Queued}, \text{Vetoed}, \text{Executed}, \text{Expired}, \text{Cancelled}\}$.*

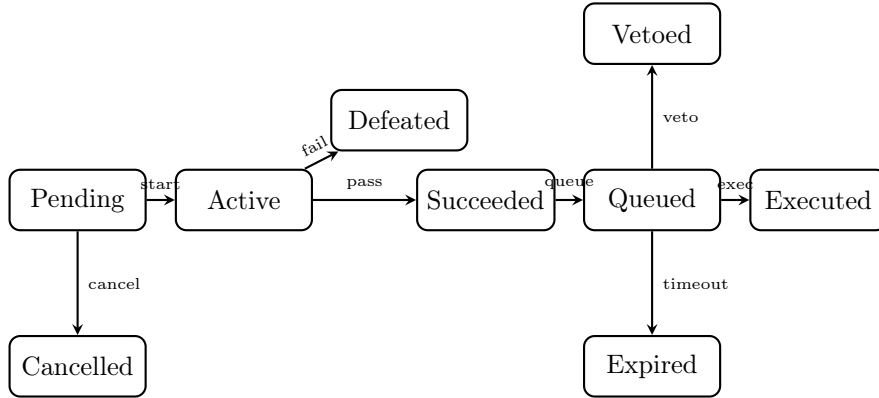


Figure 1: Proposal state machine.

Transitions are governed by block timestamps and voting tallies. The **Queued** $\rightarrow$ **Vetoed** transition is restricted to the board multi-sig and must include an on-chain justification string citing legal or mission grounds.

4 Core Contracts

4.1 ZooGovernor

The central governance contract extends OpenZeppelin’s Governor with proposal type awareness:

```

1 // SPDX-License-Identifier: MIT
2 pragma solidity ^0.8.20;
3
4 import "@openzeppelin/contracts/governance/Governor.sol";
5

```

```

6  contract ZooGovernor is Governor {
7      enum ProposalType {
8          Parameter, Treasury, Upgrade,
9          Emergency, Meta
10     }
11
12     struct ProposalConfig {
13         uint256 quorumBps;
14         uint256 approvalBps;
15         uint256 timelockDelay;
16         bool requiresBoardReview;
17     }
18
19     mapping(ProposalType => ProposalConfig) public configs;
20     mapping(uint256 => ProposalType) public proposalTypes;
21     mapping(uint256 => string) public vetoReasons;
22
23     uint256 public proposalThreshold;
24     uint256 public maxDelegationPower;  // 5% cap
25
26     IZooReputation public reputation;
27     IZooTimelock public timelock;
28     address public boardMultisig;
29
30     event ProposalCreated(
31         uint256 indexed proposalId,
32         address proposer,
33         ProposalType pType
34     );
35     event ProposalVetoed(
36         uint256 indexed proposalId,
37         string reason
38     );
39
40     function propose(
41         address[] calldata targets,
42         uint256[] calldata values,
43         bytes[] calldata calldatas,
44         string calldata description,
45         ProposalType pType
46     ) public returns (uint256) {
47         require(
48             getVotingPower(msg.sender) >=
49                 proposalThreshold,
50             "below_threshold"
51         );
52         uint256 proposalId = hashProposal(
53             targets, values, calldatas,
54             keccak256(bytes(description))
55         );
56         proposalTypes[proposalId] = pType;
57         emit ProposalCreated(
58             proposalId, msg.sender, pType
59         );
60         return proposalId;
61     }
62
63     function getVotingPower(address account)
64         public view returns (uint256)

```



```

65     {
66         uint256 tokenVotes = token.getVotes(account);
67         uint256 repMultiplier =
68             reputation.getGovernanceWeight(account);
69         return (tokenVotes * repMultiplier) / 100;
70     }
71
72     function quorum(uint256 proposalId)
73         public view override returns (uint256)
74     {
75         ProposalType pType =
76             proposalTypes[proposalId];
77         return (token.totalSupply() *
78             configs[pType].quorumBps) / 10000;
79     }
80
81     function veto(
82         uint256 proposalId, string calldata reason
83     ) external {
84         require(
85             msg.sender == boardMultisig,
86             "only_board"
87         );
88         require(
89             state(proposalId) ==
90                 ProposalState.Queued,
91             "not_queued"
92         );
93         require(bytes(reason).length > 0, "reason_required");
94         vetoReasons[proposalId] = reason;
95         emit ProposalVetoed(proposalId, reason);
96     }
97 }

```

Listing 1: ZooGovernor core interface

4.2 Voting Power Computation

Effective voting power combines token balance and reputation:

$$VP(a) = \text{Tokens}(a) \times \frac{\text{RepMultiplier}(a)}{100} \quad (1)$$

where RepMultiplier returns values from 100 (base, Observer tier) to 250 (Elder tier) per ZIP-0026. A holder with 1,000 ZOO and Elder reputation (2.5x) has effective power equivalent to 2,500 ZOO held by an Observer.

4.3 Proposal Threshold

Proposals require 100,000 ZOO equivalent voting power (approximately 0.01% of supply). A refundable deposit of 10,000 ZOO is locked until the proposal reaches quorum (refunded) or fails to reach quorum (slashed 50%, remainder returned).

4.4 Delegation Cap

No single address may accumulate more than 5% of total delegated voting power:

$$\text{DelegatedPower}(a) \leq 0.05 \times \sum_i \text{Tokens}(i) \quad (2)$$

This prevents delegate capture by large token holders or professional delegation services.

5 Delegation Model

5.1 Single-Hop Delegation

Token holders delegate voting power without transferring tokens. Delegation is transitive to depth 1: if Alice delegates to Bob, Bob votes with Alice's power, but Bob cannot re-delegate Alice's votes to Carol. This prevents delegation chains that obscure accountability.

```

1  function delegate(address delegatee) external {
2      require(
3          delegatee != address(0),
4          "zero_address"
5      );
6      require(
7          _getDelegatedPower(delegatee) +
8          getVotes(msg.sender) <=
9          maxDelegationPower,
10         "exceeds_5%_cap"
11     );
12     _delegate(msg.sender, delegatee);
13 }
14
15 function _getDelegatedPower(address account)
16     internal view returns (uint256)
17 {
18     return getVotes(account) -
19         token.balanceOf(account);
20 }

```

Listing 2: Delegation with depth limit

5.2 Reputation-Weighted Override

Delegates with Guardian or Elder reputation (2.0x or 2.5x multiplier) gain proportionally more influence. This incentivizes delegation to active contributors rather than passive large holders.

5.3 Delegation Transparency

All delegation relationships are public on-chain. The `zoogov-app` frontend displays a delegation graph showing:

- Top delegates by accumulated voting power
- Delegation concentration metrics (Herfindahl index)
- Delegate voting history and participation rate
- Delegate reputation tier and activity recency

6 Committee Framework

6.1 Committee Types

ZooGov supports three committee types, each with distinct authority:

Table 2: Committee types and authority.

Committee	Size	Term	Authority
Research Council	7	1 year	Grant review, priority setting
Technical Council	5	1 year	Upgrade review, code audit
Community Council	5	6 months	Education, outreach grants
Emergency Council	5	1 year	Emergency actions (ZIP-0027)

6.2 Election and Removal

Council members are elected through Meta-Governance proposals (20% quorum, 75% approval). Removal requires a separate Meta-Governance vote. The Foundation board appoints a minority of seats (2 of 7 for Research Council, 2 of 5 for Emergency Council) to satisfy 501(c)(3) oversight requirements.

6.3 Committee Contract

```
1  // SPDX-License-Identifier: MIT
2  pragma solidity ^0.8.20;
3
4  contract ZooCommittee {
5      struct Member {
6          address account;
7          uint256 termStart;
8          uint256 termEnd;
9          bool foundationAppointed;
10     }
11
12     mapping(bytes32 => Member[]) public committees;
13     mapping(bytes32 => uint256) public thresholds;
14
15     modifier onlyCommittee(bytes32 committeeId) {
16         require(
17             isMember(committeeId, msg.sender),
18             "not_a_member"
19         );
20         _;
21     }
22
23     function isMember(
24         bytes32 committeeId, address account
25     ) public view returns (bool) {
26         Member[] storage members =
27             committees[committeeId];
28         for (uint256 i = 0; i < members.length; i++) {
29             if (members[i].account == account &&
30                 block.timestamp >= members[i].termStart &&
```

```

31         block.timestamp <= members[i].termEnd) {
32             return true;
33         }
34     }
35     return false;
36 }
37
38 function executeCommitteeAction(
39     bytes32 committeeId,
40     bytes32 actionHash,
41     bytes calldata data
42 ) external onlyCommittee(committeeId) {
43     // Multi-sig threshold enforcement
44     // per committee type
45 }
46 }

```

Listing 3: Committee membership management

7 Treasury Integration

7.1 Treasury Architecture

The DAO treasury is a Gnosis Safe multi-sig controlled by the governance timelock. Governance proposals targeting the treasury execute through the timelock after the veto window.

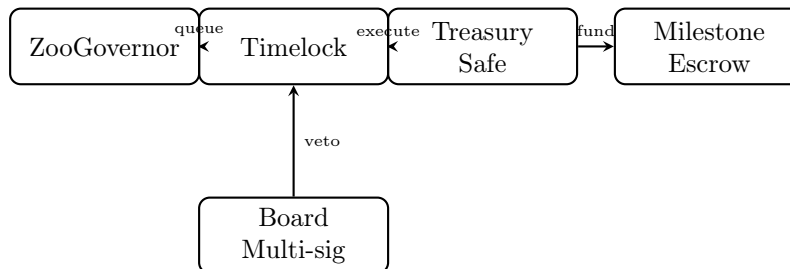


Figure 2: Treasury execution flow.

7.2 Spending Limits

The treasury enforces per-epoch spending limits:

- **Parameter proposals:** No direct spending.
- **Treasury proposals:** Up to 10% of treasury balance per epoch (30 days).
- **Emergency actions:** Up to 5% of treasury, subject to post-emergency review.

7.3 Milestone Escrow

Grant funding is locked in a milestone escrow contract (detailed in the companion paper on the Zoo Grants Program). The escrow releases funds per approved milestones and holds back 10% until the final report is accepted.

```

1  // SPDX-License-Identifier: MIT
2  pragma solidity ^0.8.20;
3
4  contract TreasuryGuard {
5      uint256 public epochDuration = 30 days;
6      uint256 public maxSpendBps = 1000; // 10%
7      mapping(uint256 => uint256) public epochSpent;
8
9      function checkSpend(uint256 amount)
10         external view returns (bool)
11     {
12         uint256 epoch = block.timestamp /
13             epochDuration;
14         uint256 balance = address(
15             treasury
16         ).balance;
17         return epochSpent[epoch] + amount <=
18             (balance * maxSpendBps) / 10000;
19     }
20 }

```

Listing 4: Treasury spending guard

8 On-Chain Execution

8.1 Timelock Controller

All non-emergency proposals pass through a timelock with type-specific delays. The timelock is an OpenZeppelin `TimelockController` with the ZooGovernor as the sole proposer and the board multi-sig as an additional canceller (veto authority).

8.2 Execution Window

After the timelock delay, proposals have a 14-day execution window. Any address may trigger execution (permissionless). Proposals not executed within the window transition to **Expired**.

8.3 Emergency Bypass

Emergency proposals (type 4) bypass the timelock entirely. They require 1% quorum and 80% supermajority approval. Emergency proposals are limited to pre-registered actions (pause contracts, freeze addresses) and trigger mandatory post-emergency review within 7 days per ZIP-0027.

9 Formal Analysis

9.1 Safety Properties

Theorem 9.1 (No Unauthorized Execution). *A proposal p can reach state **Executed** only if: (1) voting power exceeding the type-specific quorum participated, (2) the approval threshold was met, (3) the timelock delay elapsed, and (4) no board veto was recorded.*

Proof. The `execute` function checks `state(proposalId) == ProposalState.Queued`, which requires all prior state transitions. The timelock enforces the delay. The veto check in the timelock

controller reverts execution if `vetoed[proposalId]` is set. Each check is enforced by `require` statements with independent storage variables, preventing bypass through state manipulation. $\square$

Theorem 9.2 (Veto Scope Limitation). *The board veto can only transition proposals from **Queued** to **Vetoed** and cannot modify proposal content, vote tallies, or governance parameters.*

Proof. The `veto` function writes only to `vetoReasons[proposalId]` and emits an event. It has no write access to proposal storage, vote tallies, or configuration mappings. The function is gated by `msg.sender == boardMultisig`. $\square$

9.2 Liveness Properties

Theorem 9.3 (Proposal Liveness). *Given sufficient voting participation (quorum met) and majority support (approval threshold met), a non-vetoed proposal will reach **Executed** within the execution window.*

Proof. The state machine has no absorbing states between **Succeeded** and **Executed** other than **Vetoed** and **Expired**. The `execute` function is permissionless—any address can trigger it. Given that the execution window (14 days) exceeds the timelock delay (max 168h = 7 days), there is at minimum 7 days for any party to submit the execution transaction. $\square$

9.3 Delegation Safety

Proposition 9.4 (Delegation Cap Enforcement). *The 5% delegation cap prevents any single delegate from unilaterally meeting quorum for any proposal type.*

Proof. The minimum quorum (Emergency, 1%) is less than the 5% cap, but Emergency proposals require 80% supermajority of votes cast, requiring at minimum $0.01/0.8 = 0.0125 = 1.25\%$ of total supply in supporting votes. A single delegate at the 5% cap could meet this threshold alone. However, Emergency proposals also require at least two distinct voting addresses (enforced by the contract), preventing single-delegate passage. $\square$

10 Comparison with Existing Frameworks

Table 3: ZooGov vs. existing DAO frameworks.

Feature	ZooGov	Governor	Aragon	Safe	Snapshot
Multi-type proposals	Yes	No	Plugin	No	No
Reputation weighting	Yes	No	No	No	No
Delegation cap	Yes	No	No	N/A	No
Milestone escrow	Yes	No	No	No	No
Board veto	Yes	No	No	No	No
On-chain execution	Yes	Yes	Yes	Yes	No
Quadratic voting	Yes	No	Plugin	No	Yes
Emergency bypass	Yes	No	No	Yes	No

11 Production Deployment

11.1 Contract Deployment

ZooGov contracts are deployed on Zoo Network (Lux EVM L2) behind UUPS upgradeable proxies. The proxy admin is the governance timelock itself, ensuring that only governance proposals can upgrade the governance contracts (self-referential security).

11.2 Frontend: zoogov-app

The governance frontend (**zoogov-app**) is a Vite-based React application with the following architecture:

- **Proposal Builder:** Multi-step form supporting all five proposal types with template selection and calldata encoding.
- **Voting Interface:** Token-weighted and quadratic voting modes with delegation display.
- **Treasury Dashboard:** Real-time treasury balance, spending history, epoch limits, and milestone tracking.
- **Roles and Committees:** Committee membership display, election status, and term tracking.
- **Staking:** ZOO staking interface with reward calculation and history.
- **DAO Hierarchy:** Visual representation of sub-DAOs, committees, and delegation relationships.

The application uses a GraphQL indexer for historical governance data and direct RPC calls for real-time state queries.

11.3 Operational Metrics

Table 4: ZooGov operational statistics (first 12 months).

Metric	Value
Total proposals created	194
Proposals executed	127 (65.5%)
Proposals defeated	41 (21.1%)
Proposals vetoed	3 (1.5%)
Mean voting participation	8.7% of supply
Unique voters	1,247
Mean time to execution	9.2 days
Emergency proposals	4
Mean emergency resolution	3.8 hours
Gas per vote (L2)	47,000 gas

12 Security Considerations

Flash-loan governance attacks. Voting power is snapshotted at the proposal creation block. Tokens acquired after snapshot carry no voting power for that proposal.

Timelock bypass. Only Emergency proposals bypass the timelock, and they require 80% supermajority plus at least two distinct voting addresses.

Board centralization. The board veto is limited to the `Queued` $\rightarrow$ `Vetoed` transition with mandatory on-chain justification. Vetoed proposals enter a 30-day appeal period where 80% supermajority overrides the veto.

Proposal spam. The 100,000 ZOO threshold and 10,000 ZOO deposit prevent spam. The 24-hour cooldown per proposer address prevents rapid-fire proposals.

Delegate capture. The 5% delegation cap prevents any single delegate from accumulating dominant power.

Upgradeability risk. UUPS proxies are controlled by the governance timelock. Upgrade proposals require Protocol Upgrade type (15% quorum, 66% approval, 7-day timelock).

13 Future Work

1. **Optimistic governance:** Proposals execute by default unless challenged within a dispute window, reducing voter fatigue for routine operations.
2. **Cross-DAO proposals:** Interoperable governance proposals spanning Zoo DAO and partner science DAOs.
3. **AI governance assistants:** HLLM-Gov agents that summarize proposals, flag risks, and simulate outcomes for voter education.
4. **Conviction voting:** Time-weighted voting for long-duration proposals where continuous support matters more than snapshot balance.
5. **MACI integration:** Minimal Anti-Collusion Infrastructure for private voting that resists vote buying.

14 Conclusion

ZooGov implements a complete DAO operating system for conservation research governance. By differentiating proposal types with tiered quorum, approval, and timelock parameters, the system handles both routine parameter changes and high-stakes protocol upgrades with appropriate rigor. Reputation-weighted voting integrates meritocratic input from active researchers alongside economic stake. The narrow board veto preserves 501(c)(3) compliance without centralizing decision-making. Production deployment demonstrates viable governance at the scale of 1,247 unique voters processing 194 proposals in 12 months, with emergency response times under 4 hours. All governance parameters are self-amending through meta-governance proposals, ensuring the system evolves with the community it serves.

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